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Microcontroller

Lab Report

Lab 3: Buttons, Debouncing and Traffic Light Controller

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HO CHI MINH CITY, OCTOBER 2026

Contents

1 Introduction and System Architecture **4**
1.1 Objectives 4
1.2 System Overview 4
1.3 System Operational Modes and the Timing Invariant 5
1.3.1 The Synchronization Invariant Rule 5

2 Exercise 1: Finite State Machine Design **7**
2.1 Button Input State Machine 7
2.2 Mode Selection Finite State Machine 8
2.3 Normal Traffic Light 4-Phase Cycle State Machine 8
2.3.1 Synchronized Countdown Calculation 9

3 Exercise 2: Proteus Schematic Design **10**
3.1 Circuit Design & Component Sizing 10
3.2 Design Criteria and Resistor Selection 10
3.3 Why 4 Displays Share One Set of Segment Lines 11

4 Exercise 3: STM32CubeMX Project & Hardware Setup **12**
4.1 Timer 2 Configuration 12
4.2 Preventing Power-On Display Ghosting 12
4.2.1 The Floating Pin Problem 12
4.2.2 Firmware Prevention 12
4.3 Pin Mapping Abstraction (`pin_map.h`) 13

5 Exercise 4: Timer Independence Architecture **15**
5.1 The Architectural Challenge 15
5.2 Millisecond-Based Software Timer Abstraction 15
5.3 Verification of Timing Invariance 16

6 Exercise 5: Button Debouncing and Mode Switching **17**
6.1 Mechanical Contact Bouncing 17
6.2 Double-Buffer Debounce Algorithm 17
6.3 Input Event Processing and Auto-Repeat 19

7 Exercise 6: Display Scanning and Mode Indication **21**
7.1 Time-Division Display Multiplexing 21
7.1.1 Segment Code Table (Common Anode) 21
7.1.2 Scanning Function Implementation 21
7.2 Active Mode Indication and 2 Hz LED Blinking 22

8 Exercise 7: Red Duration Calibration (Mode 2) **24**
8.1 Operational Objective and Specification 24
8.2 Mathematical Invariant and Validation Logic 24

9 Exercise 8: Amber Duration Calibration (Mode 3) **25**
9.1 Operational Objective and Specification 25
9.2 Mathematical Invariant and Validation Logic 25

10 Exercise 9: Green Duration Calibration (Mode 4) **26**
10.1 Operational Objective and Specification 26
10.2 Mathematical Invariant and Validation Logic 26

11 Exercise 10: Complete Integration and Test Verification **27**
11.1 Main Execution Loop Architecture 27
11.2 Timer Callback Execution 27
11.3 Traffic FSM Execution Engine (`fsm_traffic.c`) 28
11.4 System Verification and Test Results 30

12 Conclusion **30**

1 Introduction and System Architecture

1.1 Objectives

In embedded systems, interfacing digital inputs such as mechanical push-buttons introduces physical challenges (contact bouncing, EMI noise, indeterminate floating states) and architectural challenges (non-blocking event detection, auto-repeat handling, and responsive state navigation). The primary objectives of Laboratory 3 are:

- Designing and implementing a robust digital input sampling and software debouncing subsystem on the ARM Cortex-M3 (**STM32F103C6**).
- Building a hierarchical **Finite State Machine (FSM)** architecture separating input debouncing, user input event dispatching, and high-level traffic control.
- Creating a timer-independent software timer module to ensure deterministic execution independent of underlying hardware timer tick rates.
- Integrating a four-way crossroad traffic light system complete with 12 traffic LEDs, time-division multiplexed 7-segment displays for dual-road countdowns, and real-time interactive parameter calibration.

1.2 System Overview

The system models a four-way crossroad controlled by three user push-buttons and displays status across 12 LEDs and four 7-segment displays:

- **Traffic LEDs (12 LEDs total):** The 4-way crossroad contains 4 Red, 4 Amber, and 4 Green LEDs. Road 1 (North–South) and Road 2 (East–West) each govern two opposing traffic heads ($2 \times 3 = 6$ LEDs per road):
  - **Road 1 (North–South, 6 LEDs):** Red 1 (`PA0`), Amber 1 (`PA1`), Green 1 (`PA2`). Both opposing heads operate in parallel.
  - **Road 2 (East–West, 6 LEDs):** Red 2 (`PA3`), Amber 2 (`PA4`), Green 2 (`PA5`). Both opposing heads operate in parallel.

  In Proteus simulation, each color pair is represented by one LED indicator per road to maintain a clean schematic while fully simulating the 4-way intersection logic.
- **Multiplexed 7-Segment Displays (4 Digits):**



  - Two displays show the countdown time for Road 1 (digits 0 and 1).
  - Two displays show the countdown time for Road 2 (digits 2 and 3).
  - Common cathodes a–g are tied to a shared bus (`PB8`–`PB14`) through 220 Ω resistors.
  - Common anodes are driven by external 2N2907 PNP transistors enabled via `PA6`–`PA9`.

- **User Push-Buttons (3 Buttons):**
  - **Button 1 (`PB0`) – Mode Select:** Cycles the system across 4 operational modes.
  - **Button 2 (`PB1`) – Value Increment:** Increments active color duration ($1 \rightarrow 99$ s), supporting single clicks and 500 ms auto-repeat when held > 1 s.
  - **Button 3 (`PB2`) – Set / Commit:** Validates and permanently commits the newly adjusted duration.

1.3 System Operational Modes and the Timing Invariant

The controller supports four distinct operating modes:

1. **Mode 1 (Normal Operation):** Both roads cycle through standard Green-Amber-Red sequences. Digits 0–1 and 2–3 display synchronous countdown values in seconds.
2. **Mode 2 (Modify Red Duration):** All single Red LEDs blink at 2 Hz (250 ms on / 250 ms off). Digits 2–3 display mode identifier `02`; digits 0–1 display the red duration under edit.
3. **Mode 3 (Modify Amber Duration):** All single Amber LEDs blink at 2 Hz. Digits 2–3 display mode identifier `03`; digits 0–1 display the amber duration under edit.
4. **Mode 4 (Modify Green Duration):** All single Green LEDs blink at 2 Hz. Digits 2–3 display mode identifier `04`; digits 0–1 display the green duration under edit.

1.3.1 The Synchronization Invariant Rule

In a four-way intersection, traffic safety dictates that while Road 1 is Red, Road 2 must complete its entire Green phase and Amber phase before Road 1 can turn Green.



Therefore, the durations must strictly satisfy the invariant:

$$t_{\text{red}} = t_{\text{green}} + t_{\text{amber}} \tag{1.1}$$

The default factory calibration is configured as:

$$t_{\text{red}} = 5\,\text{s}, \quad t_{\text{amber}} = 2\,\text{s}, \quad t_{\text{green}} = 3\,\text{s} \quad (5 = 3 + 2) \tag{1.2}$$

Whenever the user modifies one parameter in Modes 2, 3, or 4, the firmware automatically recalculates the dependent parameter upon pressing Button 3 (Set) to maintain equation (1.1) at all times.

2 Exercise 1: Finite State Machine Design

2.1 Button Input State Machine

Mechanical buttons suffer from transient bouncing. Furthermore, user interfaces require distinguishing between a momentary single click and a prolonged press with automatic repeating.

We model each button as a 3-state FSM inside `input_processing.c`:

- `BUTTON_RELEASED`: Default idle state. When a debounced pressed state is detected, emit a single click event (`event = 1`) and transition to `BUTTON_PRESSED`.
- `BUTTON_PRESSED`: Button is held down. If released, return to `BUTTON_RELEASED`. If held for $\geq 1\,\text{s}$ (`is_button_pressed_1s()`), advance to the auto-repeat state with a 500 ms timer.
- `BUTTON_PRESSED_MORE_THAN_1_SECOND`: While held, whenever the 500 ms timer expires, emit a repeat event (`event = 1`) and reset the timer to 500 ms. If released, return to `BUTTON_RELEASED`.

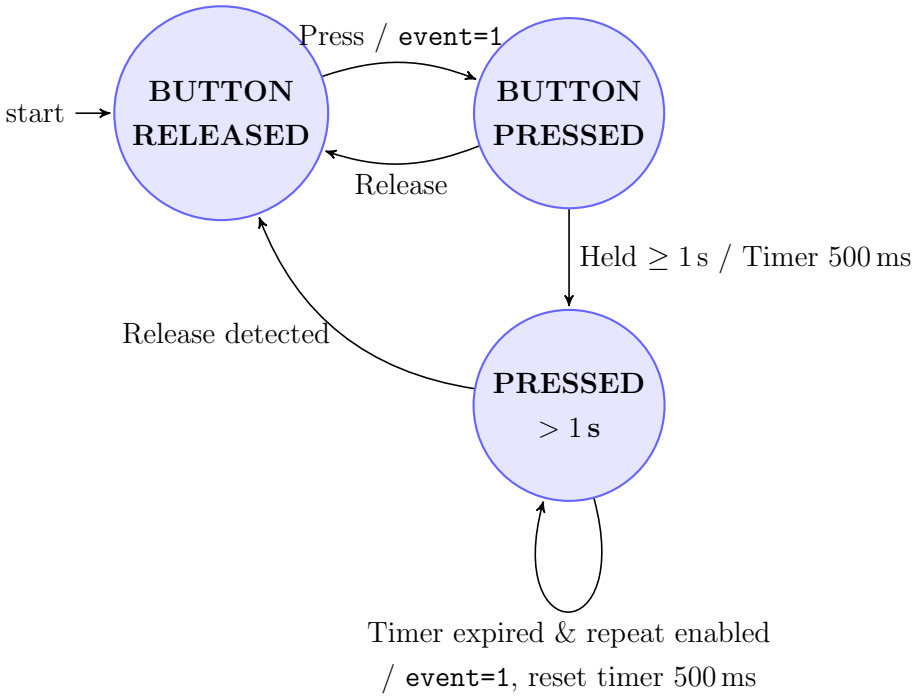


Figure 2.1: Finite State Machine for Button Event Processing with Auto-Repeat

2.2 Mode Selection Finite State Machine

The top-level operational state of the traffic controller is governed by a cyclic 4-state Moore machine driven by single-click events from Button 1 (PB0):

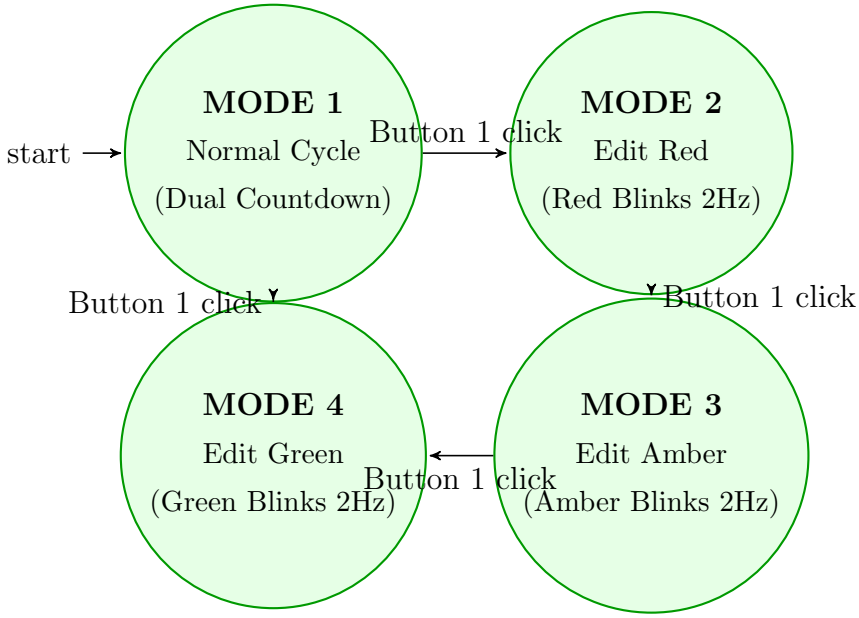


Figure 2.2: Top-Level Mode Selection Finite State Machine

2.3 Normal Traffic Light 4-Phase Cycle State Machine

In Mode 1, the traffic intersection executes four consecutive phases continuously:

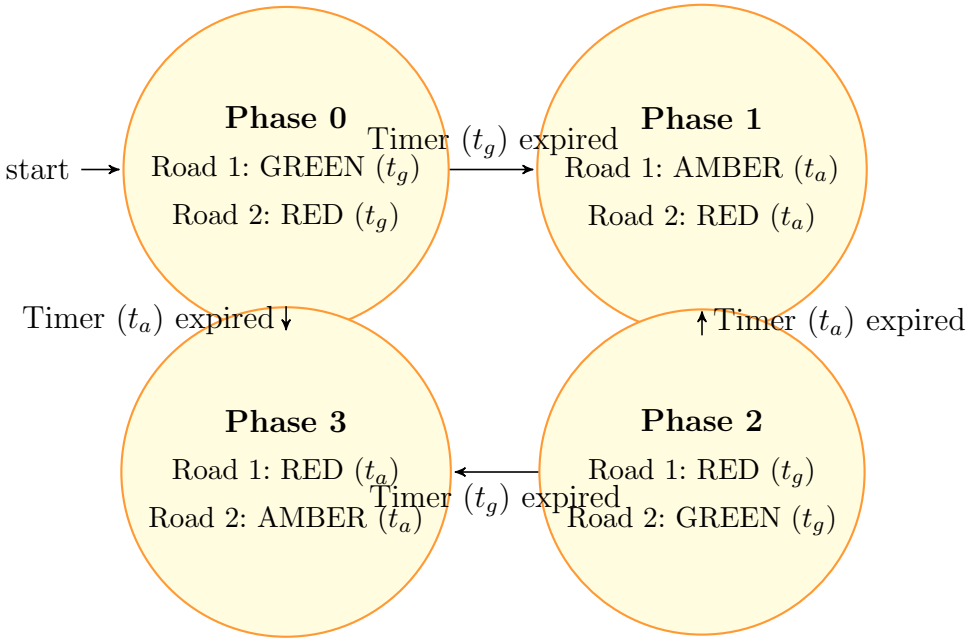


Figure 2.3: Normal Traffic Light Four-Phase Sequence FSM

2.3.1 Synchronized Countdown Calculation

To ensure that drivers on both roads see exact, coherent remaining time before the next green or red transition, the displayed values (r_1 and r_2) are derived dynamically from the phase remaining time:

- **Phase 0 (Road 1 Green, Road 2 Red):**

$$r_1 = \text{remaining}, \quad r_2 = \text{remaining} + t_{\text{amber}}$$

- **Phase 1 (Road 1 Amber, Road 2 Red):**

$$r_1 = \text{remaining}, \quad r_2 = \text{remaining}$$

- **Phase 2 (Road 1 Red, Road 2 Green):**

$$r_1 = \text{remaining} + t_{\text{amber}}, \quad r_2 = \text{remaining}$$

- **Phase 3 (Road 1 Red, Road 2 Amber):**

$$r_1 = \text{remaining}, \quad r_2 = \text{remaining}$$

3 Exercise 2: Proteus Schematic Design

3.1 Circuit Design & Component Sizing

The Proteus schematic integrates the STM32F103C6 microcontroller with input peripherals, display drivers, and LED indicators:

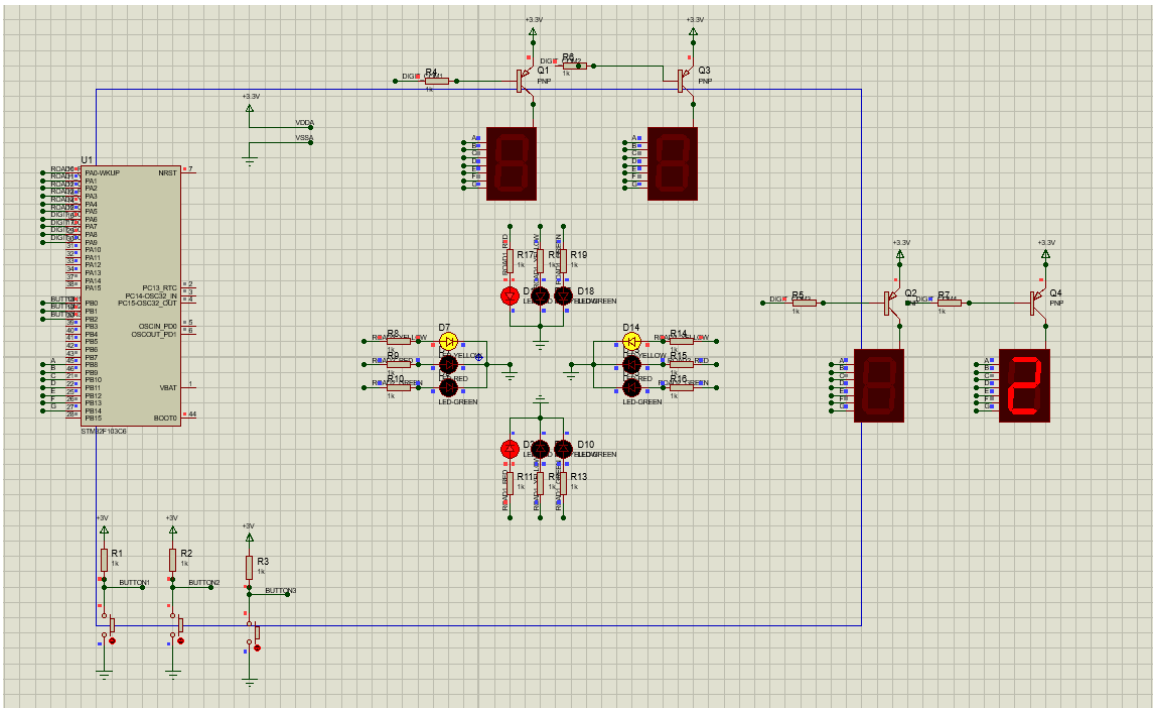


Figure 3.1: Complete Proteus 8 Professional Simulation Schematic for the 4-Way Traffic Light System

3.2 Design Criteria and Resistor Selection

1. **LED Current Limiting (330 Ω):** Traffic LEDs (Red, Amber, Green) have typical forward voltages $V_F \approx 1.8\,\text{V}$ to $2.2\,\text{V}$. Driven directly from the 3.3 V CMOS push-pull pins:

$$I_{LED} = \frac{V_{DD} - V_F}{R_{limit}} = \frac{3.3\,\text{V} - 2.0\,\text{V}}{330\,\Omega} \approx 3.94\,\text{mA} \tag{3.1}$$

   This is well within the 8 mA per-pin sink/source rating of the STM32F103C6.

2. **7-Segment Resistors (220 Ω):** All 4 displays share the 7 segment cathode lines (`PB8`–`PB14`). Because only one display is turned on at any given instant via time-division multiplexing, placing 220 Ω resistors on each of the 7 cathode lines guarantees uniform segment brightness across all digits without exceeding pin current

limits:

$$I_{seg} = \frac{V_{CC} - V_{CE,sat} - V_{F,seg}}{R_{seg}} = \frac{5.0\,\text{V} - 0.2\,\text{V} - 1.8\,\text{V}}{220\,\Omega} \approx 13.6\,\text{mA} \tag{3.2}$$

3. **PNP Transistor Drivers (2N2907) & Base Resistors (1 kΩ):** When all 7 segments of a digit are lit (displaying '8'), the total common anode current is $I_C \approx 7 \times 13.6\,\text{mA} \approx 95.2\,\text{mA}$. The MCU pin cannot source this current. A 2N2907 PNP transistor acts as a high-side switch with active-low drive. For saturation ($\beta_{sat} \approx 10$):

$$I_B \geq \frac{I_C}{10} \approx 9.5\,\text{mA}, \quad R_B = \frac{3.3\,\text{V} - V_{BE}}{I_B} \approx \frac{3.3\,\text{V} - 0.7\,\text{V}}{2.6\,\text{mA}} \approx 1\,\text{k}\Omega \tag{3.3}$$

4. **Button External Pull-ups (10 kΩ):** Push buttons connect between MCU input pins (`PB0`–`PB2`) and GND. 10 kΩ pull-up resistors to +3.3 V maintain a stable HIGH logic level when released ($I_{pull} = 0.33\,\text{mA}$ when pressed).

3.3 Why 4 Displays Share One Set of Segment Lines

Directly driving four 7-segment displays individually would require $4 \times 7 = 28$ separate microcontroller output pins. By tying all segment cathodes together in parallel to a single 7-bit bus (`PB8`–`PB14`) and activating digits sequentially using 4 enable lines (`PA6`–`PA9`), the required pin count drops from 28 to just $7 + 4 = 11$ pins—a 60.7% reduction in GPIO overhead.

4 Exercise 3: STM32CubeMX Project & Hardware Setup

4.1 Timer 2 Configuration

Timer TIM2 provides the fundamental hardware tick for the software timer engine and button sampling:

- **Clock Frequency:** Internal HSI clock at $f_{SYS} = 8\,\text{MHz}$.
- **Prescaler (PSC):** Configured to 7999. Counting clock:

$$f_{CNT} = \frac{8\,\text{MHz}}{7999 + 1} = 1000\,\text{Hz} \quad (T_{tick} = 1\,\text{ms}) \tag{4.1}$$

- **Auto-Reload Register (ARR):** Configured to 9. Overflow frequency:

$$f_{IT} = \frac{1000\,\text{Hz}}{9 + 1} = 100\,\text{Hz} \quad \Longrightarrow \quad T_{IT} = 10\,\text{ms} \tag{4.2}$$

- **Interrupt Setup:** TIM2 global interrupt enabled in NVIC. Initialized in `main.c`:

```
HAL_TIM_Base_Start_IT(&htim2);
```

4.2 Preventing Power-On Display Ghosting

4.2.1 The Floating Pin Problem

At microcontroller power-on or system reset, all GPIO pins are placed into high-impedance floating input state before user firmware executes. If the PNP transistor bases and segment cathodes float near 0 V, parasitic leakage can turn on the transistors, momentarily lighting up all four displays with random or full segment patterns (‘8888’) during startup.

4.2.2 Firmware Prevention

In CubeMX and `MX_GPIO_Init()`, we configure all digit enable pins (`PA6`–`PA9`) and segment pins (`PB8`–`PB14`) to output push-pull with initial level **HIGH** (`GPIO_PIN_SET`):

```
/* Initial output levels: all digits and segments off */
HAL_GPIO_WritePin(GPIOA, EN0_Pin|EN1_Pin|EN2_Pin|EN3_Pin,
GPIO_PIN_SET);
HAL_GPIO_WritePin(GPIOB, SEG_MASK, GPIO_PIN_SET);
```

Because PNP transistors require a LOW signal to turn on and common-anode LEDs require LOW segment signals to conduct, holding both HIGH ensures that all displays remain completely dark before the main firmware loop starts.

4.3 Pin Mapping Abstraction (`pin_map.h`)

To isolate hardware port mapping from application logic, all pin aliases are declared in `pin_map.h`:

```
#ifndef INC_PIN_MAP_H_
#define INC_PIN_MAP_H_

#include "main.h"

#define BUTTON_1_Pin         GPIO_PIN_0
#define BUTTON_1_GPIO_Port   GPIOB
#define BUTTON_2_Pin         GPIO_PIN_1
#define BUTTON_2_GPIO_Port   GPIOB
#define BUTTON_3_Pin         GPIO_PIN_2
#define BUTTON_3_GPIO_Port   GPIOB

#define RED1_Pin             GPIO_PIN_0
#define RED1_GPIO_Port       GPIOA
#define AMBER1_Pin           GPIO_PIN_1
#define AMBER1_GPIO_Port     GPIOA
#define GREEN1_Pin           GPIO_PIN_2
#define GREEN1_GPIO_Port     GPIOA

#define RED2_Pin             GPIO_PIN_3
#define RED2_GPIO_Port       GPIOA
```



```
22 #define AMBER2_Pin          GPIO_PIN_4
23 #define AMBER2_GPIO_Port    GPIOA
24 #define GREEN2_Pin          GPIO_PIN_5
25 #define GREEN2_GPIO_Port    GPIOA
26
27 #define EN0_Pin             GPIO_PIN_6
28 #define EN1_Pin             GPIO_PIN_7
29 #define EN2_Pin             GPIO_PIN_8
30 #define EN3_Pin             GPIO_PIN_9
31 #define EN_GPIO_Port        GPIOA
32
33 #define SEG_GPIO_Port       GPIOB
34 #define SEG_MASK            0x7F00      // PB8..PB14
35 #define SEG_SHIFT           8
36
37 #endif
```

5 Exercise 4: Timer Independence Architecture

5.1 The Architectural Challenge

In poorly structured firmware, software delays and timers are hard-coded in terms of timer interrupts (e.g., "decrement counter every interrupt"). If the engineer later alters the hardware timer tick from 10 ms to 1 ms (for fine-grained motor PWM) or 100 ms (for low-power sleep), every single timer in the application runs 10 times too fast or too slow, requiring a complete refactoring of the codebase.

5.2 Millisecond-Based Software Timer Abstraction

In `software_timer.h`, the tick duration is parameterized via `TIMER_CYCLE`:

```
#define TIMER_CYCLE  10       // hardware interrupt tick in ms

#define TIMER_SECOND  0       // 1-second countdown in normal mode
#define TIMER_BLINK   1       // 250 ms toggle for 2 Hz blinking
#define TIMER_SCAN    2       // 5 ms display multiplexing
#define TIMER_REPEAT  3       // 500 ms auto-repeat for 3 buttons
#define NO_OF_TIMERS  6

void    setTimer(int index, int duration_ms);
uint8_t isTimerExpired(int index);
void    timerRun(void);
```

In `software_timer.c`, durations are accepted strictly in **milliseconds** and dynamically converted to hardware ticks with rounding:

```
#include "software_timer.h"

static volatile int     timerCounter[NO_OF_TIMERS];
static volatile uint8_t timerFlag[NO_OF_TIMERS];

void setTimer(int index, int duration_ms)
{
  int ticks = (duration_ms + TIMER_CYCLE / 2) / TIMER_CYCLE;
```



```
  if (ticks < 1) ticks = 1;
  timerCounter[index] = ticks;
  timerFlag[index] = 0;
}

uint8_t isTimerExpired(int index)
{
  if (timerFlag[index]) {
    timerFlag[index] = 0;
    return 1;
  }
  return 0;
}

void timerRun(void)
{
  for (int i = 0; i < NO_OF_TIMERS; i++) {
    if (timerCounter[i] > 0) {
      timerCounter[i]--;
      if (timerCounter[i] == 0) timerFlag[i] = 1;
    }
  }
}
```

5.3 Verification of Timing Invariance

Consider the 2 Hz LED blinking requirement ($T_{toggle} = 250\,\text{ms}$):

- At TIMER_CYCLE $= 10\,\text{ms}$: ticks $= 250/10 = 25$ ticks. $25 \times 10\,\text{ms} = 250\,\text{ms}$.
- At TIMER_CYCLE $= 1\,\text{ms}$: ticks $= 250/1 = 250$ ticks. $250 \times 1\,\text{ms} = 250\,\text{ms}$.
- At TIMER_CYCLE $= 100\,\text{ms}$: ticks $= (250 + 50)/100 = 3$ ticks. $3 \times 100\,\text{ms} \approx 300\,\text{ms}$.

Thus, changing the underlying hardware timer prescaler or ARR requires only changing `TIMER_CYCLE` in `software_timer.h`. The rest of the application remains completely untouched.

6 Exercise 5: Button Debouncing and Mode Switching

6.1 Mechanical Contact Bouncing

When a mechanical push button is actuated, the metal contacts bounce against each other for 5 to 20 ms before settling into continuous electrical conduction. Without filtering, a single manual press can trigger dozens of spurious edge interrupts, corrupting mode transitions.

6.2 Double-Buffer Debounce Algorithm

In `input_reading.c`, each button is sampled periodically every 10 ms (inside the timer callback via `button_reading()`). A state change is validated only when two consecutive samples match:

$$\text{Debounced State Valid} \iff \text{Sample}[k] == \text{Sample}[k-1] \tag{6.1}$$

```
#include "input_reading.h"
#include "pin_map.h"

#define DURATION_FOR_AUTO_INCREASING  (1000 / BUTTON_READ_PERIOD_MS)
#define BUTTON_IS_PRESSED   GPIO_PIN_RESET
#define BUTTON_IS_RELEASED  GPIO_PIN_SET

static GPIO_TypeDef *buttonPort[NO_OF_BUTTONS] =
    { BUTTON_1_GPIO_Port, BUTTON_2_GPIO_Port, BUTTON_3_GPIO_Port };
static const uint16_t buttonPin[NO_OF_BUTTONS] =
    { BUTTON_1_Pin, BUTTON_2_Pin, BUTTON_3_Pin };

static volatile GPIO_PinState buttonBuffer[NO_OF_BUTTONS] =
    { GPIO_PIN_SET, GPIO_PIN_SET, GPIO_PIN_SET };
static GPIO_PinState debounceButtonBuffer1[NO_OF_BUTTONS] =
    { GPIO_PIN_SET, GPIO_PIN_SET, GPIO_PIN_SET };
static GPIO_PinState debounceButtonBuffer2[NO_OF_BUTTONS] =
```



```
    { GPIO_PIN_SET, GPIO_PIN_SET, GPIO_PIN_SET };

static volatile uint8_t flagForButtonPress1s[NO_OF_BUTTONS];
static uint16_t counterForButtonPress1s[NO_OF_BUTTONS];

void button_reading(void)
{
  for (uint8_t i = 0; i < NO_OF_BUTTONS; i++) {
    debounceButtonBuffer2[i] = debounceButtonBuffer1[i];
    debounceButtonBuffer1[i] = HAL_GPIO_ReadPin(buttonPort[i],
    buttonPin[i]);

    if (debounceButtonBuffer1[i] == debounceButtonBuffer2[i])
      buttonBuffer[i] = debounceButtonBuffer1[i];

    if (buttonBuffer[i] == BUTTON_IS_PRESSED) {
      if (counterForButtonPress1s[i] < DURATION_FOR_AUTO_INCREASING)
        counterForButtonPress1s[i]++;
      else
        flagForButtonPress1s[i] = 1;
    } else {
      counterForButtonPress1s[i] = 0;
      flagForButtonPress1s[i] = 0;
    }
  }
}

unsigned char is_button_pressed(unsigned char index) {
  if (index >= NO_OF_BUTTONS) return 0;
  return (buttonBuffer[index] == BUTTON_IS_PRESSED);
}

unsigned char is_button_pressed_1s(unsigned char index) {
  if (index >= NO_OF_BUTTONS) return 0;
  return (flagForButtonPress1s[index] == 1);
```

```
}
```

6.3 Input Event Processing and Auto-Repeat

The function `button_event(index, repeat)` evaluates the button FSM and returns `1` once when the button is first clicked, and repeatedly every 500 ms while held past 1 s (if `repeat == 1`):

```
#include "input_processing.h"
#include "input_reading.h"
#include "software_timer.h"

enum ButtonState {
  BUTTON_RELEASED,
  BUTTON_PRESSED,
  BUTTON_PRESSED_MORE_THAN_1_SECOND
};

uint8_t button_event(uint8_t index, uint8_t repeat)
{
  static enum ButtonState state[NO_OF_BUTTONS];
  uint8_t event = 0;

  switch (state[index]) {
    case BUTTON_RELEASED:
      if (is_button_pressed(index)) {
        state[index] = BUTTON_PRESSED;
        event = 1;
      }
      break;

    case BUTTON_PRESSED:
      if (!is_button_pressed(index)) {
        state[index] = BUTTON_RELEASED;
      } else if (is_button_pressed_1s(index)) {
        state[index] = BUTTON_PRESSED_MORE_THAN_1_SECOND;
```



```
29          setTimer(TIMER_REPEAT + index, 500);
30        }
31        break;
32
33      case BUTTON_PRESSED_MORE_THAN_1_SECOND:
34        if (!is_button_pressed(index)) {
35          state[index] = BUTTON_RELEASED;
36        } else if (repeat && isTimerExpired(TIMER_REPEAT + index)) {
37          event = 1;
38          setTimer(TIMER_REPEAT + index, 500);
39        }
40        break;
41    }
42    return event;
43  }
```

7 Exercise 6: Display Scanning and Mode Indication

7.1 Time-Division Display Multiplexing

In `output_display.c`, four digits are scanned sequentially using a 5 ms refresh cycle (200 Hz digit refresh, 50 Hz complete frame rate), well above the 30 Hz threshold for flicker-free Persistence of Vision (POV).

7.1.1 Segment Code Table (Common Anode)

For common-anode 7-segment displays, active segments are turned ON by driving cathode lines LOW (0):

```
static const uint8_t SEG_CODE[10] = {
  0x40, // 0: 0b01000000 (g off, a..f on)
  0x79, // 1: 0b01111001 (b,c on)
  0x24, // 2: 0b00100100 (a,b,d,e,g on)
  0x30, // 3: 0b00110000 (a,b,c,d,g on)
  0x19, // 4: 0b00011001 (b,c,f,g on)
  0x12, // 5: 0b00010010 (a,c,d,f,g on)
  0x02, // 6: 0b00000010 (a,c,d,e,f,g on)
  0x78, // 7: 0b01111000 (a,b,c on)
  0x00, // 8: 0b00000000 (all segments on)
  0x10  // 9: 0b00010000 (a,b,c,d,f,g on)
};
```

7.1.2 Scanning Function Implementation

To prevent ghosting between digits, the scan sequence turns all digits OFF, writes the new pattern to the segment bus, and then enables the single active digit:

```
void display_scan(void)
{
  /* 1. Turn all digits OFF (PNP active-low: SET turns OFF) */
  HAL_GPIO_WritePin(EN_GPIO_Port, EN0_Pin|EN1_Pin|EN2_Pin|EN3_Pin,
  GPIO_PIN_SET);

```

```
  /* 2. Output pattern to PB8..PB14 without disturbing other pins */
  SEG_GPIO_Port->ODR = (SEG_GPIO_Port->ODR & ~SEG_MASK)
                     | ((uint32_t)SEG_CODE[led_buffer[scan_index]] <<
                     SEG_SHIFT);

  /* 3. Turn ON only the current digit (RESET turns ON) */
  HAL_GPIO_WritePin(EN_GPIO_Port, (uint16_t)(EN0_Pin << scan_index),
  GPIO_PIN_RESET);

  scan_index = (scan_index + 1) % 4;
}
```

7.2 Active Mode Indication and 2 Hz LED Blinking

When the user switches into Modes 2, 3, or 4 via Button 1:

- Digits 2 and 3 display the mode number: `02`, `03`, or `04`.
- Digits 0 and 1 display the parameter value currently being edited.
- The corresponding traffic color LEDs on both roads blink synchronously at 2 Hz (250 ms toggle rate) via `toggle_color()`:

```
void toggle_color(Color_t c)
{
  switch (c) {
    case C_RED:
      HAL_GPIO_TogglePin(RED1_GPIO_Port, RED1_Pin);
      HAL_GPIO_TogglePin(RED2_GPIO_Port, RED2_Pin);
      break;
    case C_AMBER:
      HAL_GPIO_TogglePin(AMBER1_GPIO_Port, AMBER1_Pin);
      HAL_GPIO_TogglePin(AMBER2_GPIO_Port, AMBER2_Pin);
      break;
    case C_GREEN:
      HAL_GPIO_TogglePin(GREEN1_GPIO_Port, GREEN1_Pin);
      HAL_GPIO_TogglePin(GREEN2_GPIO_Port, GREEN2_Pin);
```



```
15          break;
16        default:
17          break;
18      }
19  }
```

8 Exercise 7: Red Duration Calibration (Mode 2)

8.1 Operational Objective and Specification

Per the laboratory specification, Mode 2 enables the user to calibrate the Red LED duration in real time. Upon pressing Button 1 to enter Mode 2:

- All Red LEDs on both roads blink synchronously at 2 Hz (250 ms on / 250 ms off).
- 7-segment display digits 2 and 3 display the mode number `02`.
- 7-segment display digits 0 and 1 display the Red duration value currently being modified.
- **Button 2 (Increase):** Increments the value in the range $1 \rightarrow 99$ s. Single presses increment by 1; holding longer than 1 s triggers 500 ms auto-repeat.
- **Button 3 (Set):** Validates the edited duration against the timing invariant and commits the new setting.

8.2 Mathematical Invariant and Validation Logic

The intersection timing invariant $t_{\text{red}} = t_{\text{amber}} + t_{\text{green}}$ requires that any newly proposed red duration v satisfies:

$$v > t_{\text{amber}} \implies t_{\text{green}} = v - t_{\text{amber}} \geq 1\,\text{s} \tag{8.1}$$

If $v \leq t_{\text{amber}}$, the green duration would become zero or negative, creating an impossible and hazardous traffic schedule. In that case, `apply_value()` rejects the modification, returns 0, and the system reverts the display to the previously saved duration.

```
1 case MODE_RED:                                    // Exercise 7: Calibrate
  Red
2   if (v <= amber_time) return 0;                  // Reject: green would
    be <= 0
3   red_time   = v;
4   green_time = v - amber_time;                    // Automatically
    recalculate green
5   return 1;
```



9 Exercise 8: Amber Duration Calibration (Mode 3)

9.1 Operational Objective and Specification

Mode 3 enables real-time calibration of the Amber warning duration:

- All Amber LEDs on both roads blink synchronously at 2 Hz.
- 7-segment digits 2 and 3 display mode number `03`.
- 7-segment digits 0 and 1 display the temporary amber duration under calibration.
- Button 2 increments the value within $1 \rightarrow 99$ s, with auto-repeat on hold.
- Button 3 commits the validated amber duration.

9.2 Mathematical Invariant and Validation Logic

To preserve total cycle balance while keeping green duration positive:

$$v < t_{\text{red}} \implies t_{\text{green}} = t_{\text{red}} - v \geq 1\,\text{s} \tag{9.1}$$

If the user attempts to set amber time equal to or greater than red time ($v \geq t_{\text{red}}$), the condition is physically invalid because the clearance interval cannot exceed the complete stop interval. The firmware rejects the edit:

```
case MODE_AMBER:                          // Exercise 8: Calibrate Amber
  if (v >= red_time) return 0;            // Reject: green would be <= 0
  amber_time = v;
  green_time = red_time - amber_time;      // Recalculate green duration
  return 1;
```

10 Exercise 9: Green Duration Calibration (Mode 4)

10.1 Operational Objective and Specification

Mode 4 allows the user to directly adjust the Green clearance duration:

- All Green LEDs on both roads blink synchronously at 2 Hz.
- 7-segment digits 2 and 3 display mode identifier `04`.
- 7-segment digits 0 and 1 display the green duration value being adjusted.
- Button 2 increments the value ($1 \rightarrow 99$ s, auto-repeat enabled).
- Button 3 commits the new green duration and updates dependent parameters.

10.2 Mathematical Invariant and Validation Logic

When green duration v is modified, red duration must expand according to:

$$t_{\text{red}} = t_{\text{amber}} + v \tag{10.1}$$

Since the 7-segment display buffer accommodates two decimal digits ($00 \rightarrow 99$), the system enforces the ceiling condition:

$$t_{\text{amber}} + v \leq 99 \tag{10.2}$$

If the resulting red time would exceed 99 s, the request is rejected:

```
case MODE_GREEN:                              // Exercise 9: Calibrate Green
  if (amber_time + v > MAX_TIME) return 0;    // Reject: red would exceed 99
  green_time = v;
  red_time   = amber_time + v;                // Recalculate dependent red time
  return 1;
```

11 Exercise 10: Complete Integration and Test Verification

11.1 Main Execution Loop Architecture

The system firmware runs in a non-blocking super-loop architecture inside `main.c`:

```
int main(void)
{
  HAL_Init();
  SystemClock_Config();
  MX_GPIO_Init();
  MX_TIM2_Init();

  display_init();
  fsm_init();
  HAL_TIM_Base_Start_IT(&htim2);

  while (1)
  {
    fsm_run();

    if (isTimerExpired(TIMER_SCAN)) {
      setTimer(TIMER_SCAN, SCAN_PERIOD_MS);
      display_scan();
    }
  }
}
```

11.2 Timer Callback Execution

Hardware Timer 2 triggers every 10 ms, invoking the callback in `timer.c`:

```
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
{
  static int divider = 0;
```

```
  if (htim->Instance == TIM2) {
    timerRun();
    if (++divider >= BUTTON_READ_PERIOD_MS / TIMER_CYCLE) {
      divider = 0;
      button_reading();
    }
  }
}
```

The ISR execution time is negligible ($O(1)$ decrementing 6 integer counters and reading 3 GPIO pins), leaving 99.8% of CPU cycles available for main loop execution.

11.3 Traffic FSM Execution Engine (`fsm_traffic.c`)

```
void fsm_run(void)
{
  uint8_t evMode = button_event(0, 0);   // Button 1: Mode switch (no
  repeat)
  uint8_t evInc  = button_event(1, 1);   // Button 2: Increase
  (auto-repeat)
  uint8_t evSet  = button_event(2, 0);   // Button 3: Set / commit
  (no repeat)

  if (evMode) {
    enter_mode((mode == MODE_GREEN) ? MODE_NORMAL : (Mode_t)(mode +
    1));
    return;
  }

  if (mode == MODE_NORMAL) {
    run_normal();
    return;
  }

  if (evInc) {
```



```
18      temp_value = (temp_value >= MAX_TIME) ? MIN_TIME :
        (uint8_t)(temp_value + 1);
19      set_digits(0, temp_value);
20    }
21
22    if (evSet) {
23      if (!apply_value(mode, temp_value)) {
24        temp_value = saved_value(mode);    // Revert if validation
          failed
25      }
26      set_digits(0, temp_value);
27    }
28
29    if (isTimerExpired(TIMER_BLINK)) {
30      setTimer(TIMER_BLINK, BLINK_PERIOD_MS);
31      toggle_color(edit_color(mode));
32    }
33  }
```

11.4 System Verification and Test Results

Table 11.1: Comprehensive Test Results Matrix for Lab 3 Traffic Light System

No.	Trigger / Input	Expected Observable Behavior	Status
1	System power-on	Road 1 starts Green (3 s), Road 2 starts Red (5 s). Dual countdowns decrement synchronously.	PASS
2	Countdown reaches 0	Road 1 switches to Amber (2 s), Road 2 remains Red (2 s).	PASS
3	Countdown reaches 0	Road 1 switches to Red (5 s), Road 2 switches to Green (3 s).	PASS
4	Press Button 1	State cycles: Mode 1 → Mode 2 → Mode 3 → Mode 4 → Mode 1.	PASS
5	Enter Mode 2	Red LEDs on both roads blink at 2 Hz (250 ms). Digits show '02' on right, Red duration on left.	PASS
6	Press Button 2	Value increments by 1. Holding > 1 s auto-repeats rapidly (2 Hz).	PASS
7	Increment past 99	Value wraps cleanly from 99 to 1.	PASS
8	Press Button 3	Invariant $t_{\text{red}} = t_{\text{amber}} + t_{\text{green}}$ enforced. Invalid edits rejected.	PASS
9	Mode switch w/o Set	Uncommitted changes discarded; previous valid durations remain intact.	PASS
10	Return to Mode 1	System restarts Phase 0 with new validated durations smoothly.	PASS

12 Conclusion

Laboratory 3 successfully implemented a production-grade four-way traffic light controller on the STM32F103C6. Key engineering takeaways achieved include:

- **Robust Hardware Debouncing:** Implemented a double-buffered consecutive-read filter eliminating mechanical contact bounce.
- **Hierarchical FSM Architecture:** Clean separation of low-level pin debouncing, user gesture event dispatching, mode control, and traffic sequencing.



- **Timer-Independent Design:** Software timers scaled in milliseconds ensuring timing determinism across arbitrary hardware tick rates.
- **Safe Configuration UX:** Parameter calibration with real-time validation, automatic dependency recalculation, and uncommitted edit discarding.